A dome away from home

Answers

1 January 2065

World leaders call for help

‘We are running out of room!’ claim scientists. World leaders are panicking at the thought that there is limited room left on Earth for our growing population. As a result, several of the world’s leaders have met at the World Population Summit to thrash out a solution. Just today, a spokesperson for the group has made a statement calling for assistance:

‘We are in desperate times and we need help. The members participating in the World Population Summit have decided that we have no choice but to start another human colony on Mars before it’s too late. They are calling out to all scientists, environmentalists and the like to put forward their ideas for an ecosystem on Mars that can support human life. The ideas will be trialled and then, if the trials are successful, humans will begin to live on Mars. Submissions can be made to the secretary of the World Population Crisis Committee.’

Whether we leave earth because of overpopulation, environmental concerns or just in the spirit of exploration, our new home will need to provide us with the essential ingredients to sustain human life. Wherever we go, we will need protection and shelter, a food supply, water and air to breathe.

Your task

As part of a submission to the World Population Crisis Committee, you are to design a space habitat that provides all the requirements for a trial group of 12 colonists to survive on Mars. A successful design on this small scale will then lead to the development of a much larger colony to alleviate the crowding on Earth.

You will need to consider the following as part of your submission.

(a) Water
- (i) Where will the water come from?
- (ii) What will it be used for?
- (iii) Will it be recycled?
- (iv) If it is recycled, how can it be purified so that it is drinkable again?
- (v) How much water will each person need?

(b) Food
- (i) Will the food be prepared and packaged in advance?
- (ii) Can plants for food be grown in the habitat itself? What will the plants need to survive?
- (iii) Will there be any animals used for food? What will their requirements be?
- (iv) How can plants and animals be transported to your habitat?

(c) Air
- (i) What gases are needed to support life?
- (ii) How much air will be needed for each person?
- (iii) Will the air be recycled? What methods could be used to do this?
- (iv) Where will the air come from to start with?

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(d) Shelter

(i) What will your habitat be made from?

(ii) How big will it need to be?

(iii) What temperature extremes will it need to withstand?

(iv) Are there any hazards in particular (e.g. radiation, micrometeorites, dust storms) that your habitat will need to protect your colony from?

(v) How strong will it need to be?

(vi) Will it have normal Earth gravity? Does it need normal gravity?

(e) Transport

(i) How will your colonists get to the habitat?

(ii) Will there be easy communication with Earth?

(iii) How will the colonists return to Earth?

(iv) How long will it take to get to the habitat?

Use the next two pages to write your notes on each of these aspects in the spaces provided. On the final page draw a design of your habitat, including labels that point out the important features.

WATER

Suggested answers only as responses will vary:

Water will need to be recycled and purified.

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FOOD

Suggested answers only as responses will vary:

Plants and animals will most likely be needed.

AIR

Suggested answers only as responses will vary:

The air would need to be similar to Earth air.

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SHELTER

Suggested answers only as responses will vary:

Normal gravity will be necessary for the long-term health of humans.

TRANSPORT

Suggested answers only as responses will vary:

Communication would be recommended but not easy. It is not necessary for colonists to return to Earth.

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